

Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 9/08/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

Yes. We met with Dr. Rabia Shahid on Tuesday, September 9th, at 8 p.m. (Joshua Adegboyo, Anika Baro Villa, and Brehon Parker were present). Aside from that session, no other formal team meeting occurred this week; we stayed aligned via Discord.

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

Yes. The entire team (Joshua Adegboyo, Anika Baro Villa, Nicholas Campos, Brehon Parker) met with Dr. Ali Ibrahim on Thursday, September 11th at 6 p.m.

3. Describe verbally the tasks completed the past week and the challenges faced.

This week (Week 4) we completed our work with traditional feature-selection methods, with each team member responsible for a different approach. Anika ran Sequential Feature Selector (SFS); Brehon applied Variance Threshold; Nicholas performed the chi-square (χ^2) test; and Joshua worked on LASSO regularization. For each method we created t-SNE plots to visually assess how effectively the selected features separate the cancer stages, and we compiled the selected miRNA lists to compare results and identify miRNAs that consistently appeared across the different methods. A key challenge during the feature-selection work was the high dimensionality and class imbalance of the miRNA dataset, which required careful use of class-balanced scoring and cross-validation to obtain reliable results.

Additionally, we completed the product press release that summarizes our breast cancer stage classification system for a general audience. Joshua designed the logo and drafted the contact information, dateline, and headline to establish a clear and professional opening. Anika wrote the sub-headline and introduction, focusing on presenting the system's key strengths in an engaging, easy-to-understand way. Brehon prepared the body and boilerplate, providing a concise explanation of the system's purpose and the team's background. Nicholas handled the closing section and gathered and attached photos of the team members.

4. Describe the tasks to be completed the coming week.

Next week (Week 5) our team will start using deep learning as an additional feature-selection method. Each member will design and train a different neural-network architecture, which Dr. Ali Ibrahim will assign during our upcoming meeting, on the cleaned miRNA dataset to find which miRNAs are most predictive of breast cancer stage. After training, we will examine the features flagged as important by

these deep-learning models and compare them with the subsets we identified this week using SFS, chi-square, Variance Threshold, and LASSO. This comparison would help us see how deep-learning-based feature selection complements the classical methods and whether it reveals miRNAs that consistently stand out across both approaches.

Previous Report Summary. Date: 9/01/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyoye	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

Current Report Summary. Date: 9/08/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyoye	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

Name of the Member: Joshua Adegboyo

Report Date: 9/08/25

Task 3.1: Clean miRNA expression files (9/7/25)

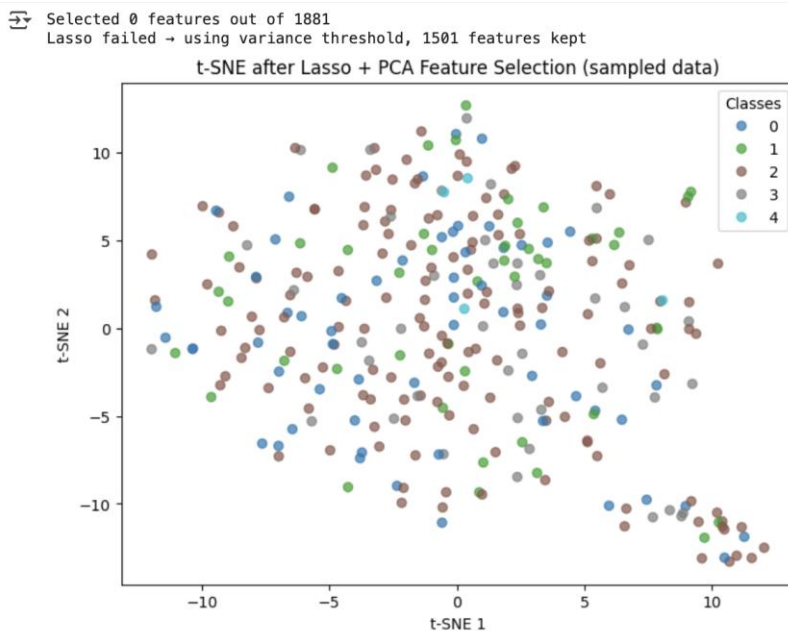
With Dr. Ali's guidance, we would begin the data cleaning procedures to ensure the raw miRNA expression files are consistent, complete, and properly normalized. This would include checking file integrity, handling missing or invalid values, applying a $\log_2(x+1)$ transformation for normalization, and removing outliers. The expected outcome would be a clean, well-structured dataset prepared for integration with clinical metadata in the next task. <https://sso.fau.edu/4p/?return=/idp/profile/SAML2/Redirect/SSO?execution=e5s1>

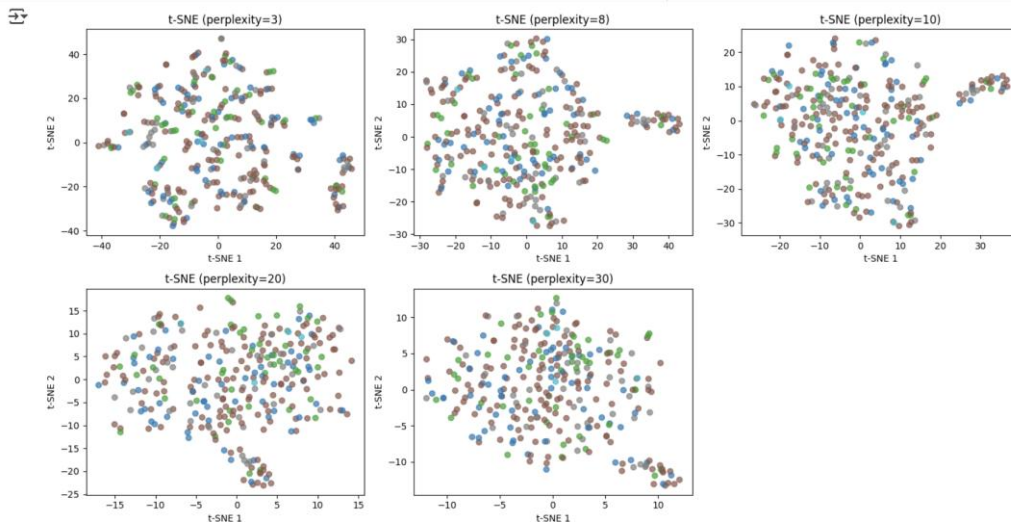
Task 3.2: Link cleaned miRNA expression data with clinical metadata by case IDs (9/7/25)

In collaboration with Dr. Ali, the team would link the cleaned miRNA expression dataset with the corresponding clinical metadata using unique case identifiers. This process would involve aligning sample IDs across both datasets, verifying the presence and accuracy of cancer stage labels (I–IV), and addressing any inconsistencies or missing entries. The team would also assess the distribution of samples across stages to understand class balance. The final output would be a fully labeled dataset, combining miRNA features with clinical information, prepared for downstream tasks such as feature selection and model training.

Task 4.1: Feature selection for relevant miRNAs identification (9/14/25)

During our meeting, Dr. Ali showed us various feature selection methods, which we would split amongst ourselves. These selections are Chi-square, NCA, and L1-norm. We will use these selections to reduce dimensionality and identify the most informative miRNAs for breast cancer stage classification. My focus was on the lasso feature selection method. Lasso feature selection was applied to the dataset to identify predictive features. However, with the chosen regularization, no features were selected.





Task 4.2: Press Release Project (**Logo, Contact information, Dateline** and the **Headline**) (9/15/25)

For this project, we were tasked to put together a product press release that would include all standard sections, such as the logo, contact, dateline, headline, sub-headline, introduction, body, boilerplate, close, and team photos. We divided the sections across each team member, and my focus was on the **Logo, Contact information, Dateline** and the **Headline**.

Task 5.1: Apply deep learning for feature selection (9/21/25)

This task will be completed collaboratively, as advised by our sponsor due to its complexity. We plan to apply deep learning methods (e.g., autoencoders or neural networks) to identify relevant features from the dataset, ensuring proper preprocessing for compatibility. The goal is to generate a reduced feature space for downstream modeling.

Task 5.2: Compare deep-learning results with traditional feature-selection methods (9/21/25)

After completing deep learning-based selection, we will compare its results with traditional methods. The comparison will focus on subset size, interpretability, performance, and stability, helping us determine the most effective strategy for later modeling phases.

Name of the Member: Anika Baro Villa

Report Date: 9/08/25

Task 3.1: Clean miRNA expression files (9/01/25)

This task involved cleaning and organizing the raw TCGA miRNA expression files using MATLAB scripts developed by Dr. Ali Ibrahim from his previous research to ensure consistency and accuracy. From each .txt file, we extracted the third column, reads_per_million_miRNA_mapped, as the normalized measure of expression for downstream analysis. We also removed annotation files that did not contain expression values, reducing the dataset from 1,244 to 1,209 usable files. The process produced two outputs: a labels struct linking file_ids to case_ids and clinical stages, and S, a matrix storing the expression values with their corresponding file_ids.

Task 3.2: Link cleaned miRNA expression data with clinical metadata by case IDs (9/01/25)

Using MATLAB scripts developed by Dr. Ali Ibrahim, we linked the cleaned expression matrix S to clinical metadata by matching file_ids to case_ids, added staging information, and standardized the labels by collapsing all sub-stages into their parent stages (for example, IA and IB were grouped into Stage I, and similarly for the other stages). After harmonization, the stages were encoded into numeric values, with 0 representing normal tissue and 1–4 representing stages I through IV. The final dataset contained 1,209 samples and 1,882 columns, with 1,881 miRNA features and one stage label column, and was exported to CSV with miRNA identifiers as headers.

Task 4.1: Apply Sequential Feature Selection (SFS) to identify the most informative miRNAs (9/12/25)

I began by reviewing scikit-learn's SFS documentation and examining the class distribution to see how many samples belonged to each stage. The snippet below shows the quick check I ran to count the samples per class. The distribution of our dataset is clearly skewed (Stage IV had very few samples) so plain accuracy would misrepresent performance.

```
# Check class balance: count how many samples belong to each class in y
unique_classes, class_counts = np.unique(y, return_counts=True)
for cls, count in zip(unique_classes, class_counts):
    print(f"Class {cls}: {count} samples")

Class 0: 265 samples
Class 1: 173 samples
Class 2: 595 samples
Class 3: 161 samples
Class 4: 15 samples
```

Based on that, I decided to evaluate each candidate feature subset by scoring the model with f1_macro so each stage counts equally, and trained the model with class_weight='balanced' so minority stages were appropriately weighted. Below are key portions of the code that show how I handled scaling, weighting, and scoring.

```
# Estimator used inside SFS (scaling inside CV; class weighting for imbalance)
pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("logreg", LogisticRegression(
        solver="lbfgs", multi_class="multinomial",
        class_weight="balanced", max_iter=20000, random_state=42
    ))
])
```

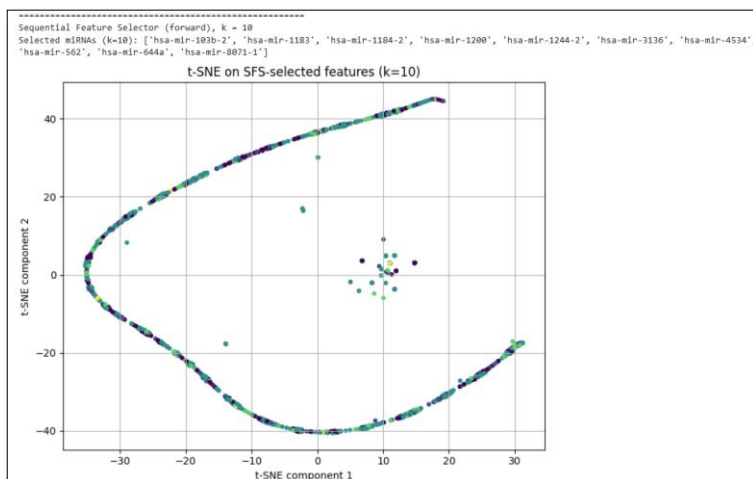
```
# Apply SFS with 3-fold CV; score macro-F1 (treats classes equally)
sfs = SequentialFeatureSelector(
    estimator=pipe,
    n_features_to_select=k,
    direction="forward",
    scoring="f1_macro",
    cv=3,
    n_jobs=-1
)
```

As the screenshots above show, to avoid data leakage, I decided to wrap preprocessing and the classifier in a single scikit-learn Pipeline (StandardScaler followed by multinomial LogisticRegression) and used that pipeline as the estimator inside SFS. This way, the scaler is fit only on the training portion of each cross-validation fold. I ran stratified 3-fold cross-validation ($cv=3$) to balance runtime and stability, and used forward selection, which starts with no features and adds the best feature at each step until the target subset size k is reached.

As I progressed, I had to manage runtime. SFS proved to be the most computationally demanding feature selection method our team tested, since at each step it re-fits the model for every remaining feature across the cross-validation folds. On our high-dimensional matrix, the first full pass took hours. To keep it tractable, and following Dr. Ali Ibrahim's instructions, I limited the runs to the predefined subset sizes $k = 10, 8, 5$, and 3 , kept $cv=3$, used the same lightweight LR pipeline, enabled parallelism ($n_jobs=-1$), and kept the workflow minimal (fit, save the list, plot t-SNE, then compute the overlap once at the end).

For each k I fit SFS, I saved the selected miRNA list to a CSV so the team could compare lists across k and check overlaps, and plotted a t-SNE using only the selected, re-scaled features to visually assess separability. In the end I produced four per- k gene lists and four t-SNE plots, plus a concise overlap summary highlighting miRNAs that recur across k . The t-SNE plots show a smooth, ribbon-like manifold with mixed labels (indicating limited separability from very small subsets) while the overlap analysis provides a stable shortlist for the team to compare across methods. A small core of three miRNAs appears in every list, and five additional miRNAs appear in at least two lists. The screenshots below show the saved CSV files, an example t-SNE plot for $k = 10$, and the overlap summary.

sfs_k3_features	9/12/2025 7:07 PM	Microsoft Excel C...	1 KB
sfs_k5_features	9/12/2025 7:05 PM	Microsoft Excel C...	1 KB
sfs_k8_features	9/12/2025 7:01 PM	Microsoft Excel C...	1 KB
sfs_k10_features	9/12/2025 6:55 PM	Microsoft Excel C...	1 KB



Overlap summary ($k = 10, 8, 5, 3$):

Common to ALL ($10 \cap 8 \cap 5 \cap 3$): ['hsa-mir-1244-2', 'hsa-mir-3136', 'hsa-mir-8071-1']

Appear in ≥ 2 of $\{10, 8, 5, 3\}$: ['hsa-mir-103b-2', 'hsa-mir-1183', 'hsa-mir-1244-2', 'hsa-mir-3136', 'hsa-mir-4534', 'hsa-mir-562', 'hsa-mir-644a', 'hsa-mir-8071-1']

Task 4.2: Draft a product press release (sub-headline and introduction) (9/14/25)

For this task, I worked on writing the sub-headline and introduction of our press release. I thought about how to present the system in a way that shows its main strengths and makes people curious to read more. The sub-headline was meant to get across the core benefit quickly, while the introduction pulled together the key details in a clear and engaging way. Additionally, I reviewed the sections my teammates drafted, offering suggestions to improve the document and made overall formatting changes to keep the press release consistent and compliant with the requirements.

Task 5.1: Apply deep learning for feature selection (9/21/25)

Next week I plan to design and train the specific neural-network architecture that Dr. Ali Ibrahim will assign to me during our upcoming meeting, while the other team members work on their own designs. Using our cleaned and labeled miRNA dataset, I will tune key settings such as the number of layers, neurons, activation functions, and regularization, so the model captures meaningful patterns without overfitting. My goal is to use the trained network's weights and interpretability tools to identify which miRNAs are most relevant for predicting breast cancer stage.

Task 5.2: Compare deep-learning results with traditional feature-selection methods (9/21/25)

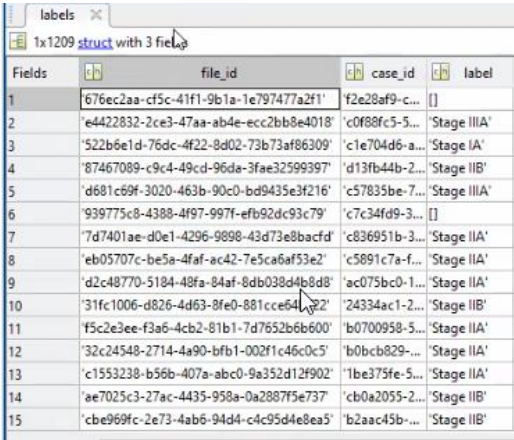
After training my assigned neural-network architecture, I plan to examine the miRNAs it highlights as most important and compare those with the feature subsets we produced this week using traditional machine-learning methods. I will look for overlaps and also note any unique markers that emerge only from the deep-learning model. My goal is to identify miRNAs that consistently stand out across both approaches, giving a clearer picture of which features are the strongest candidates for classification.

Name of the Member: **Nicholas Campos**

Report Date: **9/08/25**

Task 3.1: Cleaning data (9/1/25)

Under Dr. Ali's direction, our team will implement comprehensive data preprocessing protocols to ensure miRNA expression files meet quality standards for analysis. This process will involve validating file formats and completeness, addressing missing or corrupted data points, implementing $\log_2(x+1)$ normalization transformations, and identifying and removing statistical outliers. The deliverable will be a standardized, analysis-ready dataset suitable for clinical metadata integration.



Fields	file_id	case_id	label
1	'676ec2aa-cf5c-41f1-9b1a-1e797477a2f1'	'f2e28af9-c...	['']
2	'e4422832-2ce3-47aa-ab4e-ecc2bb8e4018'	'c0f88fc5-5...	'Stage IIA'
3	'522b6e1d-75dc-4f22-8d02-73b73af86309'	'c1e704d6-a...	'Stage IA'
4	'87467089-c9c4-49cd-96da-3fae32599397'	'd13fb44b-2...	'Stage IIB'
5	'd681c69f-3020-463b-90c0-bd9435e3f216'	'c57835be-7...	'Stage IIIA'
6	'939775c8-4388-4f97-997f-efb92dc93c79'	'c7c34fd9-3...	['']
7	'7d7401ae-d0e1-4296-9898-43d73e8bacfd'	'c836951b-3...	'Stage IIA'
8	'eb05707c-be5a-4faf-ac42-7e5ca6af53e2'	'c5891c7a-f...	'Stage IIA'
9	'd2c48770-5184-48fa-84af-8db038d4b8d8'	'ac075bc0-1...	'Stage IIA'
10	'31fc1006-d826-4d63-8fe0-881cce64a22'	'24334ac1-2...	'Stage IIB'
11	'f5c2e3ee-f3a6-4cb2-81b1-7d7652b6b600'	'b0700958-5...	'Stage IIA'
12	'32c24548-2714-4a90-bfb1-002f1c46c0c5'	'b0bcb829-...	'Stage IIA'
13	'c1553238-b56b-407a-abc0-9a352d12f902'	'1be375fe-5...	'Stage IIA'
14	'ae7025c3-27ac-4435-958a-0a2887f5e737'	'cb0a2055-2...	'Stage IIB'
15	'cbe969fc-2e73-4ab6-94d4-c4c95d4e8ea5'	'b2aac45b-...	'Stage IIB'

Task 3.2: Using MATLAB combine labels with data (9/1/25)

Our sponsor Dr. Ali Ibrahim gave us a walkthrough and then used his MATLAB scripts to combine the cleaned miRNA expression data with clinical stage labels. The code matched case identifiers between datasets to align samples with their corresponding cancer stage classifications (I-IV). This created a unified labeled dataset ready for feature selection and model training phases.

Task 4.1: Chi2 Feature Selection Implementation and Analysis (9/11/25)

I successfully implemented Chi-squared feature selection using a systematic loop-based approach to evaluate 10, 8, and 3 features for breast cancer stage classification. Using scikit-learn's SelectKBest with chi2 scoring function, I applied a for loop to iterate through each feature count (for k in [10, 8, 3]), ensuring consistent preprocessing with data normalization to handle the non-negative requirement for Chi2 testing. For each iteration, I applied the feature selection, transformed the data, and generated t-SNE visualizations to assess clustering patterns across cancer stages. The analysis revealed that 10 features provided the best visual separation between cancer stages compared to 8 and 3 features.


```

# Feature selection (Chi2, try 10 features, 8 features, 3 features and plot it using tsne
from sklearn.feature_selection import SelectKBest, chi2

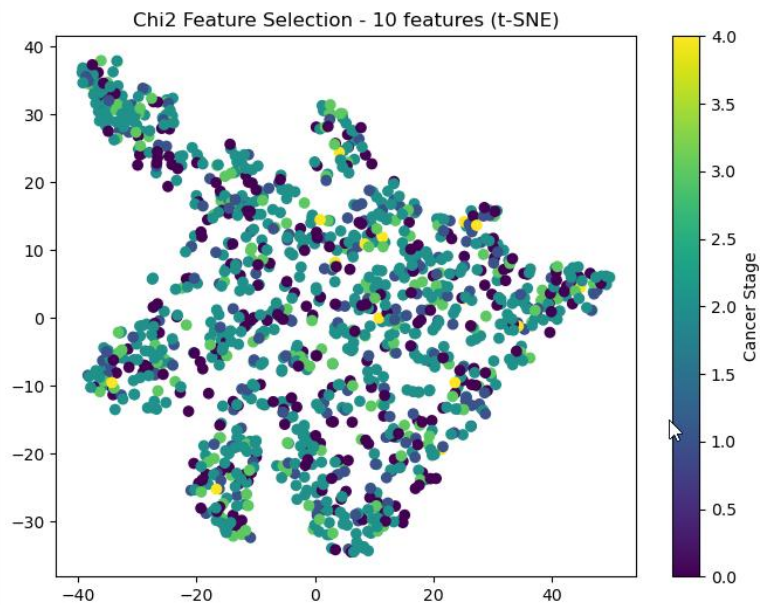
# Make data non-negative for chi2
data_positive = data - data.min() + 1e-10

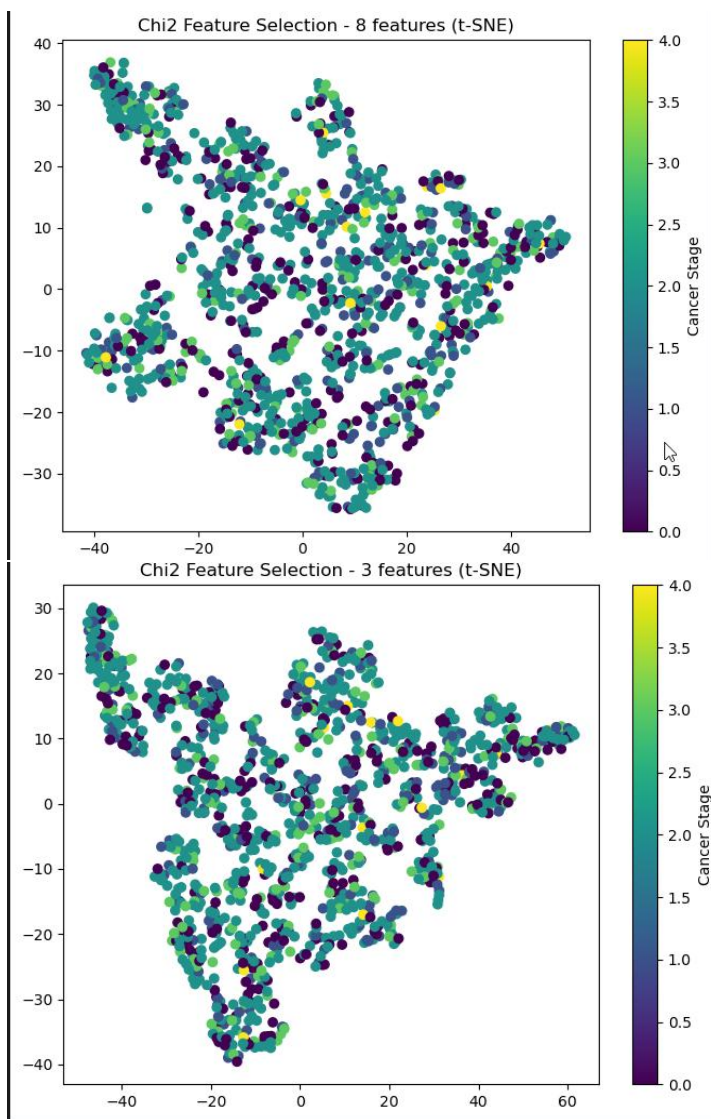
# Chi2 feature selection with 10, 8, 3 features
for k in [10, 8, 3]:
    # Apply Chi2 feature selection
    chi2_selector = SelectKBest(score_func=chi2, k=k)
    data_selected = chi2_selector.fit_transform(data_positive, label)

    # Apply t-SNE
    tsne = TSNE(n_components=2)
    tsne_result = tsne.fit_transform(data_selected)

    # Plot t-SNE
    plt.figure(figsize=(8, 6))
    plt.scatter(tsne_result[:, 0], tsne_result[:, 1], c=label, cmap='viridis')
    plt.title(f'Chi2 Feature Selection - {k} features (t-SNE)')
    plt.colorbar(label='Cancer Stage')
    plt.show()

```





Task 4.2: Product Press Release Finalization and Team Photos Integration (9/11/25)

I developed the comprehensive "Key Features" bullet points section highlighting our system's technical capabilities including advanced miRNA expression analysis, comparative ML/DL evaluation, rigorous data splitting protocols, and multiple feature selection methods. I also created the "Meet the Team" page with professional photos and names for all team members using proper formatting and layout design.

Task 5.1: Apply deep learning for feature selection (9/18/25)

Next week I'll implement deep learning-based feature selection approaches to complement my traditional Chi2 method from the previous week. I'll develop neural network architectures that can automatically identify important miRNA features through techniques such as attention mechanisms, layer-wise relevance propagation, or gradient-based feature importance scoring. Using the same dataset and 70/15/15 split from my Chi2 analysis, I'll train deep learning models and extract feature importance rankings. I'll use a similar loop-based approach to evaluate different numbers of top features (10, 8, 3) for consistency with my Chi2 implementation. The deliverable will be trained deep learning models with documented feature importance scores and selected miRNA biomarkers ready for comparative analysis.

Task 5.2: Traditional vs Deep Learning Feature Selection Comparative Analysis (9/18/25)

Building on both my Chi2 implementation and deep learning feature selection from the previous week, I'll create a systematic comparison framework evaluating both methods across multiple criteria: computational efficiency, feature stability across different runs, biological interpretability of selected features, and downstream classification performance. The analysis will include overlap studies between Chi2-selected features and deep learning-identified features, examining whether both approaches converge on similar miRNA biomarkers or reveal complementary information. I'll quantify agreements and differences between the methods and assess which provides more reliable biomarker identification. The deliverable will be a comprehensive comparative report with recommendations on traditional vs deep learning feature selection approaches for breast cancer stage classification, informing our team's overall ML vs DL comparison strategy.

Name of the Member: Brehon Parker

Report Date: 9/08/25

Task 3.1:

I coordinated the Zoom meeting with Dr. Ali Ibrahim and the team to review the preprocessing workflow for the TCGA miRNA expression files. This included scheduling the time, creating and sharing the Zoom link, and confirming attendance for all group members. During the session, Dr. Ali walked us through how the MATLAB scripts extract the third column (*reads_per_million_miRNA_mapped*) from each file, why this normalization is used, and how annotation files without expression values should be removed. I helped facilitate the discussion by asking clarifying questions to ensure our team understood how the file IDs mapped to expression data. I also documented these instructions, which the group later used to guide the cleaning process and reduce the dataset from 1,244 to 1,209 usable files. (9/7/25)

Task 3.2:

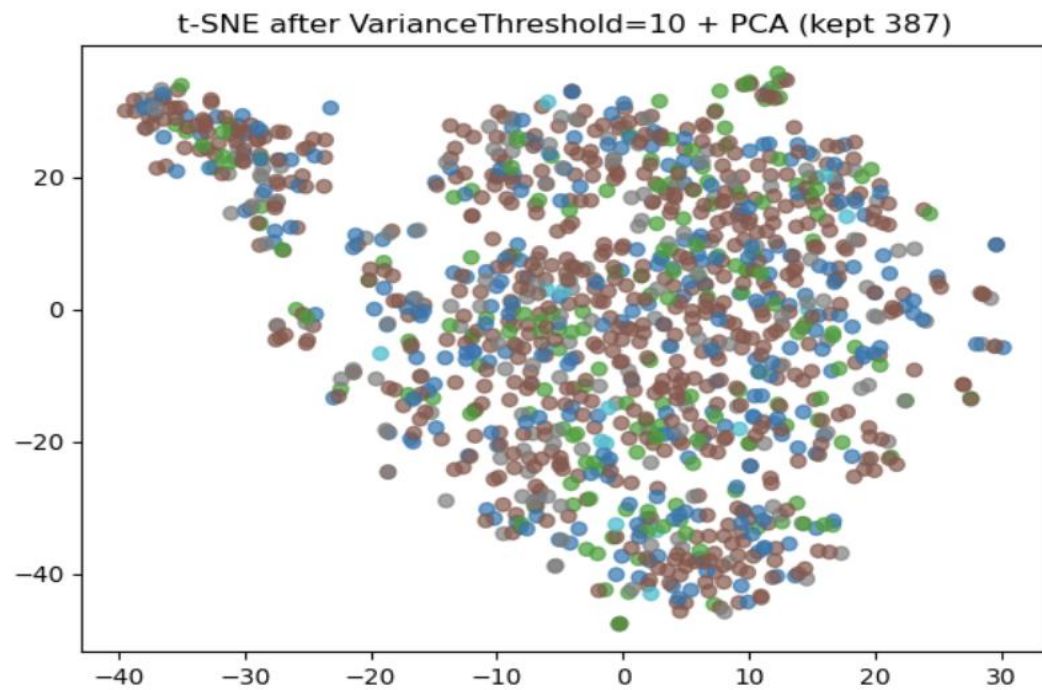
For this part of the project, we focused on linking the cleaned miRNA expression files with the patient clinical metadata. I set up the meeting with Dr. Ali Ibrahim by scheduling the time, sharing the Zoom link, and making sure everyone could attend. In the session, Dr. Ali walked us through how the *file_id* values from the expression data should be matched to the *case_id* entries in the clinical files so that each patient's data lined up correctly. He also explained how the staging information needed to be simplified by combining sub-stages (like IA and IB into Stage I) and then assigning numeric codes, with 0 for normal and 1–4 for cancer stages I through IV. I worked alongside my teammates as we followed these steps, which gave us a merged dataset of 1,209 samples and 1,882 features, now including the encoded stage labels. This step was important because it tied the expression data to clinical context, giving us a dataset we can actually use for classification in the next phase of the project. (9/7/25)

Task 4.2 Deep learning for feature selection implementation (Variance Threshold) (9/11/25)

I applied the VarianceThreshold method at thresholds of 10, 8, and 3 to reduce the original 1,881 features. This resulted in keeping 387, 402, and 459 features respectively. After dimensionality reduction, PCA was applied to stabilize the variance, and the results were visualized using t-SNE. The figures below show the t-SNE embeddings for each threshold, with color representing cancer stage labels. These reduced datasets were also exported for reproducibility. This step advanced the project by narrowing the feature space while maintaining separation across labels, preparing the data for downstream modeling.

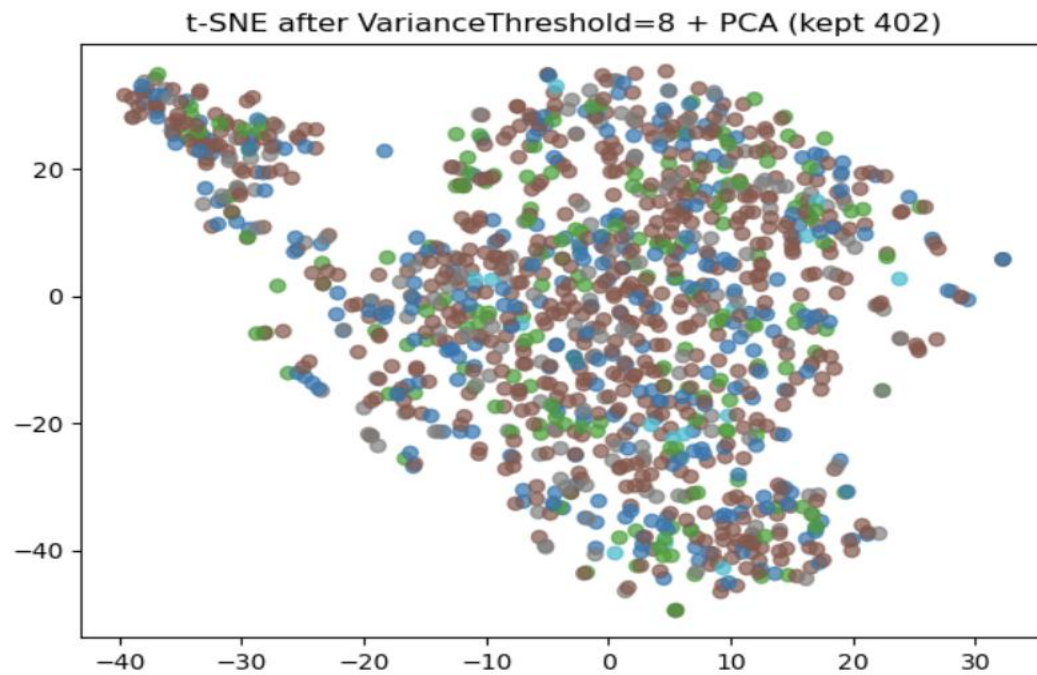
VT=10 (kept 387 features out of 1,881)

[VT=10] kept 387 / 1881 features

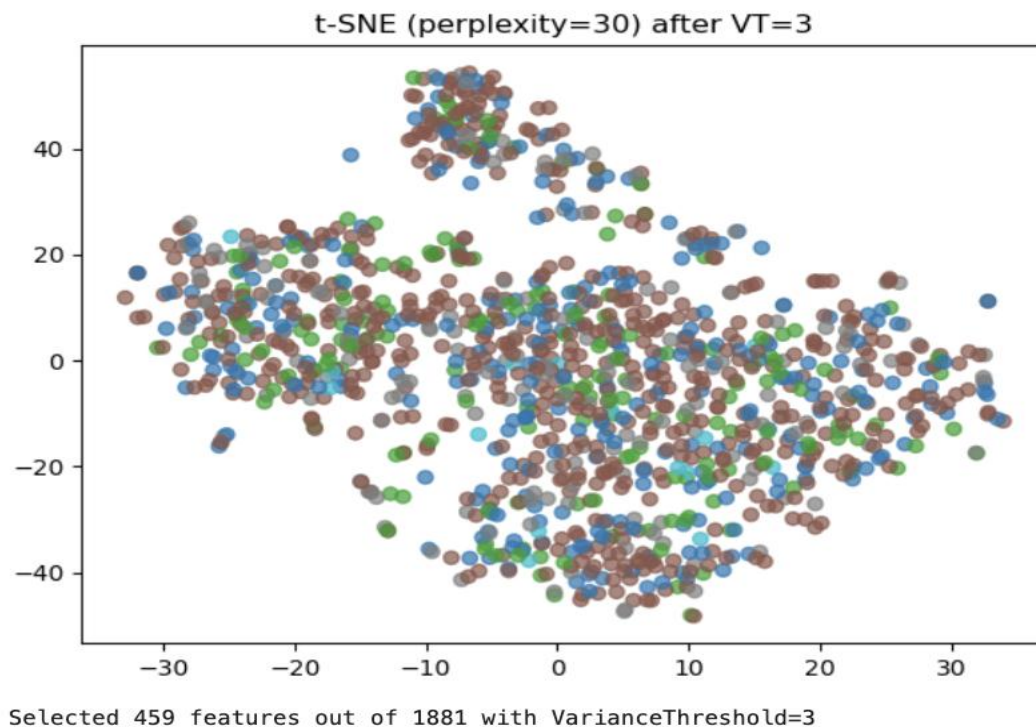


VT=8 (kept 402 features out of 1,881)

[VT=8] kept 402 / 1881 features



VT=3 (kept 459 features out of 1,881)



Task 4.3 Product Press Release – Body/Boilerplate Sections (9/14/25)

We worked on a product press release that follows the usual structure, with sections like the logo, contact details, dateline, headline, sub-headline, introduction, body, boilerplate, closing, and team photos. Each of us was responsible for different parts, and my task was to draft the **body and boilerplate**. In the body, I focused on clearly explaining what our system does, its main features, and the value it brings. For the boilerplate, I put together background information about our team and the project so readers have context on who we are and what we're building.

Task 5.1: Apply deep learning for feature selection (9/21/25)

For this task, we will be working with Dr. Ali to apply deep learning methods to our dataset. With this task requiring a heavier load, it was agreed that we will work as group under Dr. Ali's direction to complete this task as a group. Our main focus will be on setting up the model and identifying important features for classification.

Task 5.2: Compare deep learning results with traditional feature-selection methods (9/21/25)

For this task, we will work with Dr. Ali to compare the features selected by deep learning with the results from all traditional methods already applied: SequentialFeatureSelector, VarianceThreshold, Chi², and Lasso. The goal is to measure overlap, highlight differences, and determine which features are most stable and informative for downstream modeling.